Exp No: 9 Experiment to understand Matplotlib library use cases in Data Science through visualization.

AIM:

To conduct an experiment to understand Matplotlib library use cases in Data Science through visualization.

PROGRAM:

```
import pandas as pd

import numpy as np

import seaborn as sns

import matplotlib.pyplot as plt

%matplotlib inline

data = pd.read_csv('/content/drive/MyDrive/Iris_Dataset.csv')

data

data.info()

data.describe()

data.value_counts('variety')

sns.countplot(x='variety', data=data,)

plt.show()

dummies = pd.get_dummies(data.variety)

FinalDataset = pd.concat([pd.get_dummies(data.variety), data.iloc[:, [0, 1, 2, 3]]], axis=1)

FinalDataset.head()

sns.scatterplot(x='SepalLengthCm', y='SepalWidthCm', hue='variety', data=data,)
```

```
sns.scatterplot(x='PetalLengthCm', y='PetalWidthCm', hue='variety',
data=data,)

sns.pairplot(data, hue='variety', height=3);

plt.show()

sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot,
'PetalLengthCm').add_legend();

plt.show();

sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot,
'PetalWidthCm').add_legend();

plt.show();

sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot,
'SepalLengthCm').add_legend();

plt.show();

sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot,
'SepalWidthCm').add_legend();

plt.show();
```

OUTPUT:

```
<class 'pandas.core.frame.DataFrame'>

RangeIndex: 150 entries, 0 to 149

Data columns (total 6 columns):

 #   Column        Non-Null Count  Dtype

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 0   Id            150 non-null    int64

 1   SepalLengthCm  150 non-null    float64

 2   SepalWidthCm   150 non-null    float64

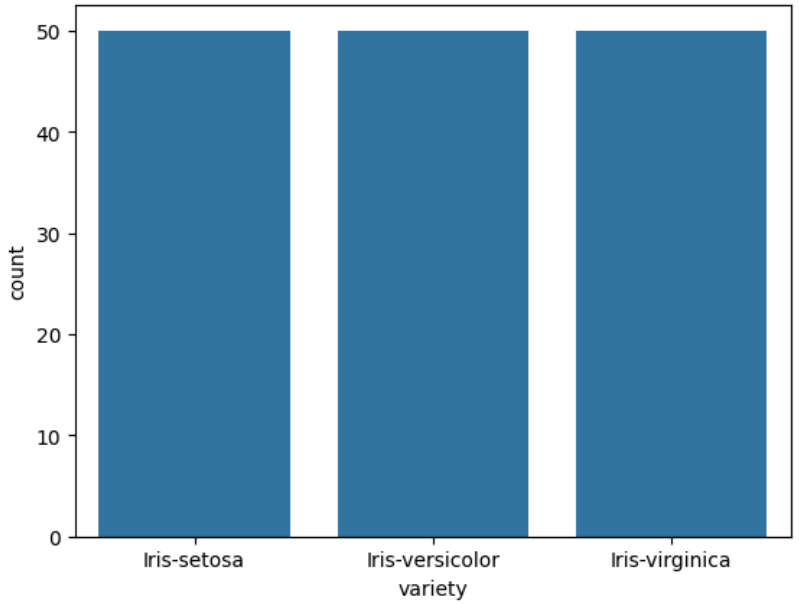
 3   PetalLengthCm  150 non-null    float64
```

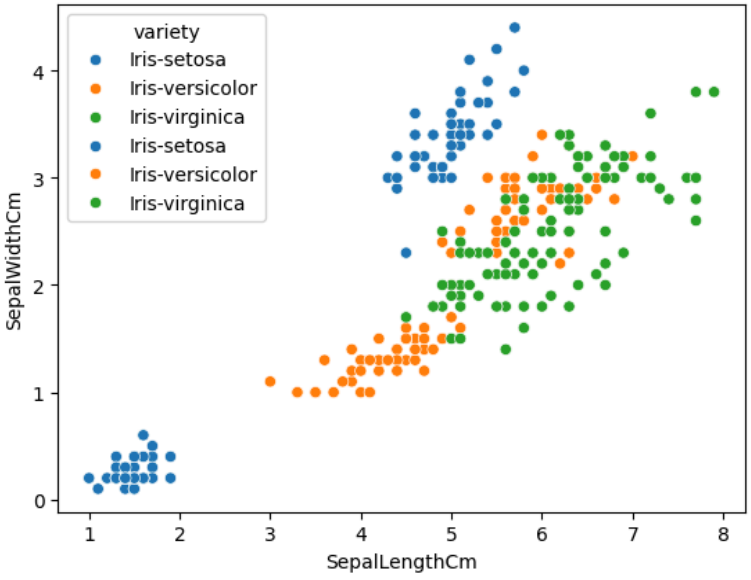
4 PetalWidthCm 150 non-null float64

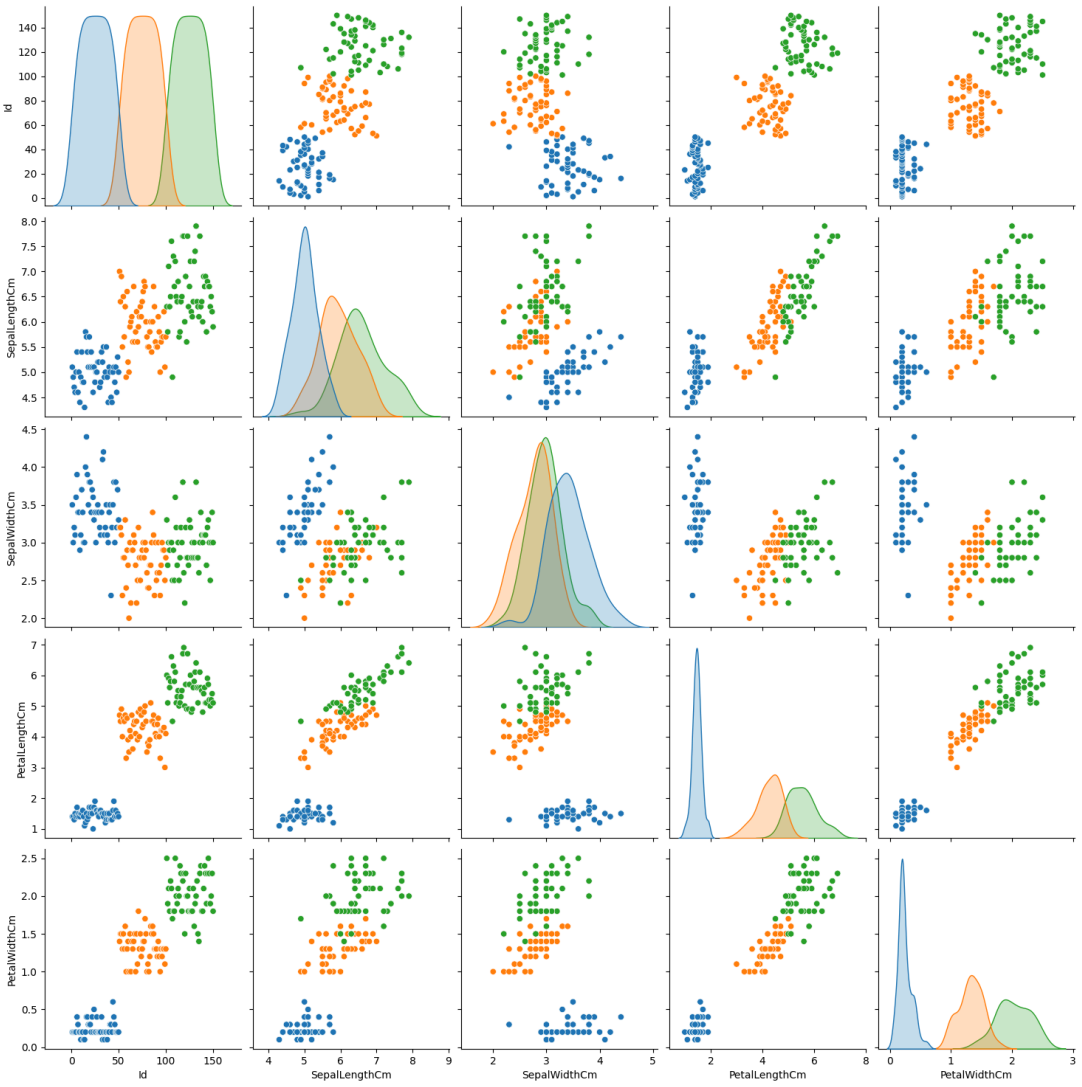
5 variety 150 non-null object

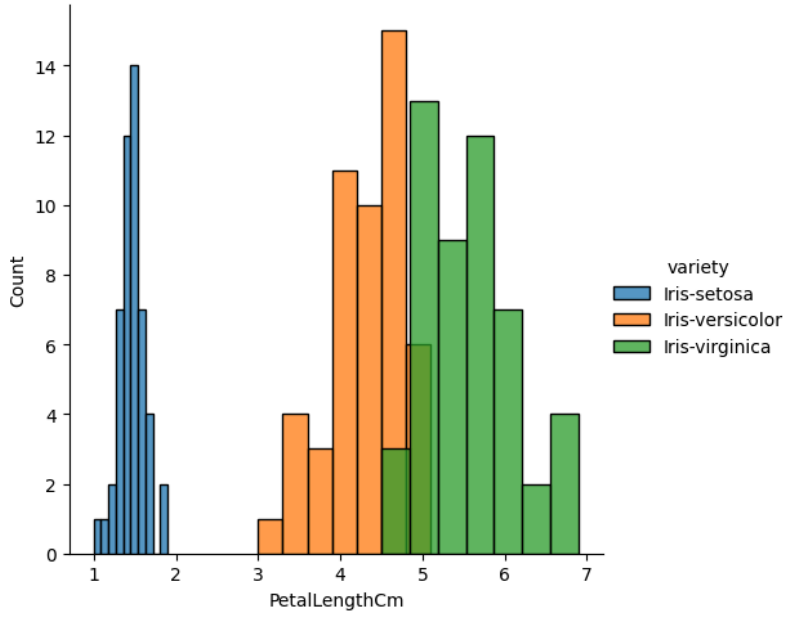
dtypes: float64(4), int64(1), object(1)

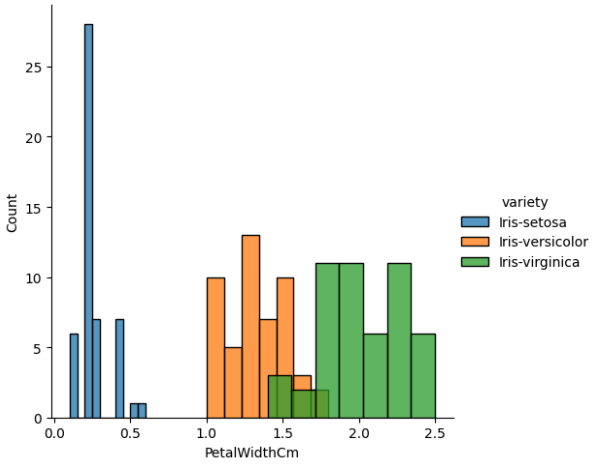
memory usage: 7.2+ KB

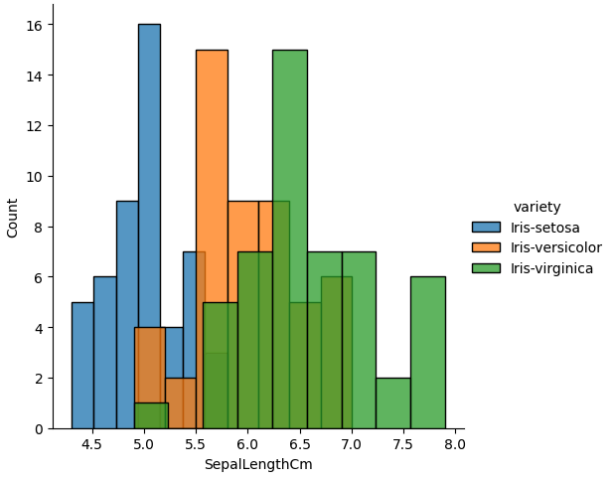


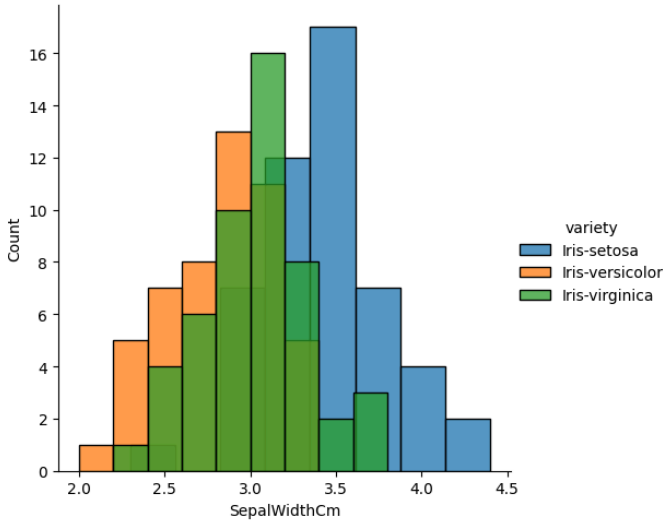



Id
SepalLengthCm
SepalWidthCm
PetalLengthCm
PetalWidthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica


Count
PetalLengthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica







RESULT:

Thus an experiment to understand Matplotlib library use cases in Data Science through visualization was conducted.